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Chứng minh ngắn công thức xtime (trong GF(2^n)):
  g(x) = g_(n-1)x^(n-1) + ... + g_0; m(x) = x^n + r(x), deg(r) < n.
  xg(x) = g_(n-1)x^n + [các hạng tử bậc < n].
  Vì m(x) = 0 (mod m), nên x^n = r(x) (mod m); trong GF(2), trừ chính là XOR.
  Suy ra xtime(g,m) = ((g << 1) & MASK) XOR [g_(n-1) * (m & MASK)], MASK = 2^n - 1.
  g_(n-1)=0: chỉ dịch trái; g_(n-1)=1: bỏ bit tràn rồi XOR phần thấp của m.

Ví dụ theo đúng vòng lặp multiply của gf2m4.py: f=1011, g=0110, m=10011.
  Khởi tạo result=0000, term=0110. Xét f_i từ bit thấp (i=0) đến bit cao (i=3):
  i  f_i  term     cập nhật result                  term cho vòng kế tiếp
  0   1   0110     0000 ^ 0110 = 0110              xtime(0110) = 1100
  1   1   1100     0110 ^ 1100 = 1010              11000 ^ 10011 = 01011 -> 1011
  2   0   1011     giữ 1010                       10110 ^ 10011 = 00101 -> 0101
  3   1   0101     1010 ^ 0101 = 1111              không tính xtime nữa
  Kết quả: f(x)*g(x) mod m(x) = 1111 = x^3 + x^2 + x + 1.

01  GF(2^4) XOR      1010 ^ 0111 = 1101
02  GF(2^4) XOR      1111 ^ 1111 = 0000
03  GF(2^4) XOR      0101 ^ 0011 = 0110
04  GF(2^4) XOR      0000 ^ 1011 = 1011
05  GF(2^4) XTIME    g=0000, m=10011:  0000 << 1 = 00000 -> 0000
06  GF(2^4) XTIME    g=0001, m=10011:  0001 << 1 = 00010 -> 0010
07  GF(2^4) XTIME    g=0101, m=10011:  0101 << 1 = 01010 -> 1010
08  GF(2^4) XTIME    g=1010, m=10011:  1010 << 1 = 10100; 10100 ^ 10011 = 00111 -> 0111
09  GF(2^4) XTIME    g=1000, m=10011:  1000 << 1 = 10000; 10000 ^ 10011 = 00011 -> 0011
10  GF(2^4) XTIME    g=1000, m=11001:  1000 << 1 = 10000; 10000 ^ 11001 = 01001 -> 1001
11  GF(2^4) XTIME    g=1000, m=11111:  1000 << 1 = 10000; 10000 ^ 11111 = 01111 -> 1111
12  GF(2^4) MULTIPLY f=0000, g=1011, m=10011;  bit f=1 tai i=khong co
      P = 0 = 0000000
      P bac < 4  ->  R = 0000
13  GF(2^4) MULTIPLY f=0001, g=1011, m=10011;  bit f=1 tai i=0
      P = 0001011 = 0001011
      P bac < 4  ->  R = 1011
14  GF(2^4) MULTIPLY f=1011, g=0101, m=10011;  bit f=1 tai i=0,1,3
      P = 0000101 ^ 0001010 ^ 0101000 = 0100111
      R1: 0100111 ^ 0100110 (m<<1) = 0000001  ->  0001
15  GF(2^4) MULTIPLY f=1011, g=0110, m=10011;  bit f=1 tai i=0,1,3
      P = 0000110 ^ 0001100 ^ 0110000 = 0111010
      R1: 0111010 ^ 0100110 (m<<1) = 0011100
      R2: 0011100 ^ 0010011 (m<<0) = 0001111  ->  1111
16  GF(2^4) MULTIPLY f=0010, g=1000, m=11001;  bit f=1 tai i=1
      P = 0010000 = 0010000
      R1: 0010000 ^ 0011001 (m<<0) = 0001001  ->  1001
17  GF(2^4) MULTIPLY f=0010, g=1000, m=11111;  bit f=1 tai i=1
      P = 0010000 = 0010000
      R1: 0010000 ^ 0011111 (m<<0) = 0001111  ->  1111
18  GF(2^4) MULTIPLY f=1111, g=1111, m=10011;  bit f=1 tai i=0,1,2,3
      P = 0001111 ^ 0011110 ^ 0111100 ^ 1111000 = 1010101
      R1: 1010101 ^ 1001100 (m<<2) = 0011001
      R2: 0011001 ^ 0010011 (m<<0) = 0001010  ->  1010
19  GF(2^8) XOR      01010111 ^ 10000011 = 11010100
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20  GF(2^8) XOR      11111111 ^ 11111111 = 00000000
21  GF(2^8) XOR      00000000 ^ 10101010 = 10101010
22  GF(2^8) XOR      10100000 ^ 00001111 = 10101111
23  GF(2^8) XTIME    g=00000000, m=100011011:  00000000 << 1 = 000000000 -> 00000000
24  GF(2^8) XTIME    g=00000001, m=100011011:  00000001 << 1 = 000000010 -> 00000010
25  GF(2^8) XTIME    g=01010111, m=100011011:  01010111 << 1 = 010101110 -> 10101110
26  GF(2^8) XTIME    g=10101110, m=100011011:  10101110 << 1 = 101011100; 101011100 ^ 100011011 = 001000111 -> 01000111
27  GF(2^8) XTIME    g=01000111, m=100011011:  01000111 << 1 = 010001110 -> 10001110
28  GF(2^8) XTIME    g=10001110, m=100011011:  10001110 << 1 = 100011100; 100011100 ^ 100011011 = 000000111 -> 00000111
29  GF(2^8) XTIME    g=10000000, m=100011011:  10000000 << 1 = 100000000; 100000000 ^ 100011011 = 000011011 -> 00011011
30  GF(2^8) XTIME    g=10000000, m=100011101:  10000000 << 1 = 100000000; 100000000 ^ 100011101 = 000011101 -> 00011101
31  GF(2^8) MULTIPLY f=00000000, g=10000011, m=100011011;  bit f=1 tai i=khong co
      P = 0 = 000000000000000
      P bac < 8  ->  R = 00000000
32  GF(2^8) MULTIPLY f=00000001, g=10000011, m=100011011;  bit f=1 tai i=0
      P = 000000010000011 = 000000010000011
      P bac < 8  ->  R = 10000011
33  GF(2^8) MULTIPLY f=01010111, g=00010011, m=100011011;  bit f=1 tai i=0,1,2,4,6
      P = 000000000010011 ^ 000000000100110 ^ 000000001001100 ^ 000000100110000 ^ 000010011000000 = 000010110001001
      R1: 000010110001001 ^ 000010001101100 (m<<2) = 000000111100101
      R2: 000000111100101 ^ 000000100011011 (m<<0) = 000000011111110  ->  11111110
34  GF(2^8) MULTIPLY f=01010111, g=10000011, m=100011011;  bit f=1 tai i=0,1,2,4,6
      P = 000000010000011 ^ 000000100000110 ^ 000001000001100 ^ 000100000110000 ^ 010000011000000 = 010101101111001
      R1: 010101101111001 ^ 010001101100000 (m<<5) = 000100000011001
      R2: 000100000011001 ^ 000100011011000 (m<<3) = 000000011000001  ->  11000001
35  GF(2^8) MULTIPLY f=00000010, g=10000000, m=100011011;  bit f=1 tai i=1
      P = 000000100000000 = 000000100000000
      R1: 000000100000000 ^ 000000100011011 (m<<0) = 000000000011011  ->  00011011
36  GF(2^8) MULTIPLY f=00000010, g=10000000, m=100011101;  bit f=1 tai i=1
      P = 000000100000000 = 000000100000000
      R1: 000000100000000 ^ 000000100011101 (m<<0) = 000000000011101  ->  00011101
37  GF(2^8) MULTIPLY f=01010011, g=11001010, m=100011011;  bit f=1 tai i=0,1,4,6
      P = 000000011001010 ^ 000000110010100 ^ 000110010100000 ^ 011001010000000 = 011111101111110
      R1: 011111101111110 ^ 010001101100000 (m<<5) = 001110000011110
      R2: 001110000011110 ^ 001000110110000 (m<<4) = 000110110101110
      R3: 000110110101110 ^ 000100011011000 (m<<3) = 000010101110110
      R4: 000010101110110 ^ 000010001101100 (m<<2) = 000000100011010
      R5: 000000100011010 ^ 000000100011011 (m<<0) = 000000000000001  ->  00000001
38  GF(2^4) FULL     f=1011, g=0101, m=10011;  bit f=1 tai i=0,1,3
      P = 0000101 ^ 0001010 ^ 0101000 = 0100111
      R1: 0100111 ^ 0100110 (m<<1) = 0000001  ->  0001
39  GF(2^4) FULL     f=1011, g=0110, m=11001;  bit f=1 tai i=0,1,3
      P = 0000110 ^ 0001100 ^ 0110000 = 0111010
      R1: 0111010 ^ 0110010 (m<<1) = 0001000  ->  1000
40  GF(2^8) FULL     f=01010111, g=10000011, m=100011011;  bit f=1 tai i=0,1,2,4,6
      P = 000000010000011 ^ 000000100000110 ^ 000001000001100 ^ 000100000110000 ^ 010000011000000 = 010101101111001
      R1: 010101101111001 ^ 010001101100000 (m<<5) = 000100000011001
      R2: 000100000011001 ^ 000100011011000 (m<<3) = 000000011000001  ->  11000001
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